<u>Recall</u>: • $\sum_{n=1}^{\infty} a_n$ converges (to s) $\iff$ $s_n = a_1 + \dots + a_n$ converges (to s)

• $\sum_{n=1}^{\infty} \frac{1}{n^\alpha}$ converges $\iff$ $\alpha > 1$.

<u>Rmk</u> Warning: The exact values of $\sum_{n=1}^{\infty} \frac{1}{n^\alpha}$ are not easy to compute. E.g.,

- $\alpha = 2$: $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$
- $\alpha = 4$: $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$
- $\alpha = 3$: No known formula

<u>Thm</u> (The dyadic criterion) Let $\{a_n\}_{n\geq 1}$ be a decreasing sequence with $a_n \geq 0$ $\forall n \geq 1$. The series $\sum_{n=1}^{\infty} a_n$ converges if and only if the series $\sum_{n=0}^{\infty} 2^n a_{2^n}$ converges.

<u>Pf</u>: For $n \geq 1$, set $s_n = \sum_{k=1}^{n} a_k = a_1 + a_2 + \dots + a_n$

For $n \geq 0$, set $t_n = \sum_{k=0}^{n} 2^k a_{2^k} = a_1 + 2a_2 + 4a_4 + \dots + 2^n a_{2^n}$

As $a_n \geq 0$ $\forall n \geq 1$, both $\{s_n\}_{n\geq 1}$ and $\{t_n\}_{n\geq 0}$ are increasing. So:

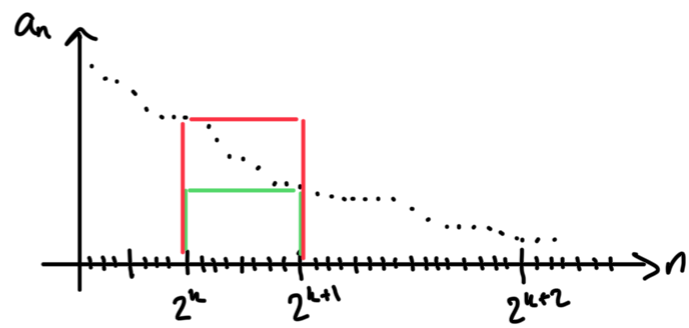
$\sum a_n$ converges $\iff$ $\{s_n\}_{n\geq 1}$ converges $\iff$ $\{s_n\}_{n\geq 1}$ bounded

$\sum 2^n a_n$ converges $\iff$ $\{t_n\}_{n\geq 0}$ converges $\iff$ $\{t_n\}_{n\geq 0}$ bounded.

So it suffices to prove:

$\{s_n\}_{n\geq 1}$ bounded $\iff$ $\{t_n\}_{n\geq 0}$ bounded.

Consider $\displaystyle\sum_{\ell=2^k+1}^{2^{k+1}} a_\ell$.



As $\{a_n\}_{n\geq 1}$ is decreasing, we have

$$\underline{\frac{1}{2}\cdot 2^{k+1} a_{2^{k+1}}} = 2^k a_{2^{k+1}} \leq \sum_{\ell=2^k+1}^{2^{k+1}} a_\ell \leq 2^k a_{2^k+1} \leq \underline{2^k a_{2^k}} \quad \forall k$$

$$\Rightarrow \underbrace{\frac{1}{2}\sum_{k=0}^{n} 2^{k+1} a_{2^{k+1}}}_{=\frac{1}{2}(t_{n+1}-a_1)} \leq \underbrace{\sum_{k=0}^{n}\sum_{\ell=2^k+1}^{2^{k+1}} a_\ell}_{\substack{= a_2+(a_3+a_4)+(a_5+\dots+a_8) \\ +\dots+(\quad\dots+a_{2^{n+1}}) \\ = s_{2^{n+1}}-a_1}} \leq \underbrace{\sum_{k=0}^{n} 2^k a_{2^k}}_{=t_n}$$

$$\Rightarrow \begin{cases} \frac{1}{2}(t_{n+1}-a_1) \leq s_{2^{n+1}}-a_1 & \Rightarrow \quad t_{n+1} \leq 2s_{n+1}-a_1 \\ s_{2^{n+1}}-a_1 \leq t_n & \Rightarrow \quad s_n \underset{\substack{\uparrow \\ n\leq 2^{n+1}\ \forall n\geq 1}}{\leq} s_{2^{n+1}} \leq t_n + a_1 \end{cases}$$

Altogether:

$\{s_n\}_{n\geq 1}$ bounded $\Rightarrow$ $\exists M>0$ s.t. $|s_n|\leq M$ $\forall n\geq 1$

$\Rightarrow$ $0 \leq t_{n+1} \leq 2M + a_1$ $\forall n\geq 1$

$\{t_n\}_{n\geq 0}$ bounded $\Rightarrow$ $\exists L>0$ s.t. $|t_n|\leq L$ $\forall n\geq 0$

$\Rightarrow$ $0 \leq s_n \leq L + a_1$ $\forall n\geq 0$

So $\{s_n\}_{n\geq 1}$ bounded $\iff$ $\{t_n\}_{n\geq 0}$ bounded. $\square$

Thm (Root test) Let $\sum_{n=1}^{\infty} a_n$ be a series.

① If $\limsup_{n\to\infty} |a_n|^{\frac{1}{n}} < 1$, then $\sum_{n=1}^{\infty} a_n$ converges absolutely.

② If $\liminf_{n\to\infty} |a_n|^{\frac{1}{n}} > 1$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Rmk If $\liminf_{n\to\infty} |a_n|^{\frac{1}{n}} \le 1 \le \limsup_{n\to\infty} |a_n|^{\frac{1}{n}}$, then the root test is inconclusive. E.g.,

- $\sum \frac{1}{n}$ diverges, and $\lim_{n\to\infty} |a_n|^{\frac{1}{n}} = \lim_{n\to\infty} \frac{1}{\sqrt[n]{n}} = 1$ Lecture 10
- $\sum \frac{1}{n^2}$ converges, and $\lim_{n\to\infty} |a_n|^{\frac{1}{n}} = \lim_{n\to\infty} \left(\frac{1}{\sqrt[n]{n}}\right)^2 = 1$

Pf: ① Set $L = \limsup_{n\to\infty} |a_n|^{\frac{1}{n}} < 1$,

$\varepsilon = \frac{1-L}{2}$ so that $L < L+\varepsilon < 1$.

Then

$$L+\varepsilon > L = \lim_{N\to\infty} \sup_{n\ge N} |a_n|^{\frac{1}{n}}$$

$$\Rightarrow \exists N_0 \text{ s.t. } \sup_{n\ge N_0} |a_n|^{\frac{1}{n}} < L+\varepsilon$$

$$\Rightarrow |a_n|^{\frac{1}{n}} < L+\varepsilon \quad \forall n \ge N_0$$

$$\Rightarrow |a_n| < (L+\varepsilon)^n \quad \forall n \ge N_0.$$

As $L+\varepsilon < 1$, then the geometric series

$$\sum_{n=N_0}^{\infty} (L+\varepsilon)^n = (L+\varepsilon)^{N_0} \sum_{k=0}^{\infty} (L+\varepsilon)^k = (L+\varepsilon)^{N_0} \frac{1}{1-(L+\varepsilon)}$$

converges (by Lecture 11). So, by the comparison test, $\sum_{n=N_0}^{\infty} |a_n|$ converges, and thus

$$\sum_{k=1}^{n} |a_k| = |a_1| + \dots + |a_{N_0-1}| + \sum_{k=N_0}^{n} |a_k|$$

converges as $n \to \infty$ as well.

(2) $1 < \liminf_{n\to\infty} |a_n|^{\frac{1}{n}} = \lim_{N\to\infty} \inf_{n\geq N} |a_n|^{\frac{1}{n}}$

$\Rightarrow \exists N_0$ s.t. $\inf_{n\geq N_0} |a_n|^{\frac{1}{n}} > 1$

$\Rightarrow |a_n|^{\frac{1}{n}} > 1 \quad \forall n \geq N_0 \quad \Rightarrow \quad |a_n| > 1 \quad \forall n \geq N_0$

$\Rightarrow a_n \not\to 0$ as $n \to \infty \quad \Rightarrow \quad \sum a_n$ diverges. $\square$